Predicting Urban OD Matrices from Synthetic Traffic Data Using an MLP: Application to Manhattan

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Abstract

Origin–destination (OD) matrices are key to transport planning, traffic management, and digital-twin applications, but are still mainly derived from household travel surveys and direct observations, which are costly and infrequently updated. An alternative is to infer OD demand indirectly from link traffic counts, yet this leads to an ill-posed inverse problem, particularly under sparse or unreliable detector coverage. This paper proposes a supervised deep-learning framework that reconstructs OD matrices from zonal traffic counts derived from link flows on a simulated urban network.

Link volumes are aggregated into intra-zonal and adjacent-zone flows by summing internal and boundary links between K-means zones. These sparse zonal counts, augmented with origin and destination totals and intra-zonal flows, form a compact input to a Multi-Layer Perceptron (MLP) that predicts full OD matrices. The training framework combines a composite loss function that embeds flow-conservation properties with systematic hyperparameter optimization based on Bayesian optimization using a Tree-structured Parzen Estimator (TPE) sampler and Hyperband pruning implemented in Optuna.

On a 20-zone Manhattan network, the final tuned model achieves strong reconstruction accuracy, indicating that both the spatial structure and global magnitude of demand can be recovered from incomplete observations. A sensitivity analysis where link flows are randomly removed before zonal aggregation shows that performance remains relatively stable under moderate data loss but deteriorates under extreme sparsity. These results suggest that hyperparameter-optimized zonal neural estimators can serve as practical surrogates for OD estimation under limited and noisy traffic measurements.

Keywords: Origin–Destination Matrix Estimation, Deep Learning, Hyperparameter Optimization, Traffic Count Data, Multilayer Perceptron

1 Introduction

Origin–destination (OD) matrices are a core input to transport planning, traffic management, and emerging digital-twin applications. They describe how trips are distributed between zones and underpin assignment, congestion analysis, and real-time control. Traditionally, OD matrices are obtained from household travel surveys and other direct observation campaigns (e.g., interviews,

license plate or GPS surveys). While methodologically well established, these approaches are expensive, time-consuming, and typically updated only at multi-year intervals, which limits their suitability for operational traffic management. As a result, there is growing interest in indirect OD estimation methods that infer demand from routinely collected data such as link traffic counts. However, reconstructing OD matrices from link flows is an ill-posed inverse problem, since many different OD patterns can reproduce the same set of observed volumes, especially under sparse or incomplete sensor coverage. The challenge is to reconstruct a plausible OD matrix from limited traffic data in a way that is accurate, computationally efficient, and robust to missing information.

Classical OD estimation methods address this problem via optimization and statistical modeling. Gravity-type and entropy-based formulations Robillard (1975); Van Zuylen and Willumsen (1980); Bell (1983), Bayesian and Poisson-likelihood models Maher (1983); Spiess (1987); Hazelton (2000), and generalized least-squares estimators Cascetta (1984); Bell (1991) combine prior matrices, traffic counts, and network loading under explicit assumptions on behavior and error structure. Bilevel and dynamic formulations further incorporate congestion, time dependence, and sensor placement Yang et al. (1994); Yang and Zhou (1998); Cascetta et al. (1993); Ashok and Ben-Akiva (2000); Zhou and Mahmassani (2006). While theoretically well-founded, these approaches are demanding to calibrate, depend on strong modeling assumptions (e.g., user equilibrium, reliable priors), and can be difficult to deploy in large networks or in real time. More recently, neural-network and deep-learning models have emerged as data-driven alternatives that approximate directly the nonlinear mapping between traffic counts and OD flows Kim and Chang (2008); Mussone and Matteucci (2013); Pamuła and Żochowska (2023); Min et al. (2025), but they often rely on high-dimensional link-flow inputs and do not systematically enforce conservation constraints or assess robustness to sensor data loss.

At the same time, the performance of neural networks is strongly influenced by the choice of hyperparameters that govern architecture, regularization, and training dynamics, such as layer width, dropout rate, learning rate, batch size, and weight decay Bergstra and Bengio (2012); Snoek et al. (2012); Probst et al. (2019). In high-dimensional search spaces, manual tuning and grid search become inefficient, while sequential model-based optimization and multi-fidelity strategies offer a more suitable framework for expensive deep-learning workloads Bergstra et al. (2011); Li et al. (2018a); Falkner et al. (2018). This is especially relevant in the present problem, where the OD estimator depends not only on network architecture and optimization settings but also on interacting loss terms introduced to preserve spatial structure and flow consistency.

This paper proposes a supervised deep-learning framework that reconstructs OD matrices from sparse zonal traffic counts derived from link flows in a simulated Manhattan network. Link volumes are first aggregated into a compact zonal representation, which is then used as input to an MLP trained with a composite loss that distinguishes OD shape from total volume and incorporates basic flow-conservation properties. To improve model selection in this high-dimensional and computationally expensive setting, the training framework incorporates systematic hyperparameter optimization using Bayesian optimization with a Tree-structured Parzen Estimator (TPE) sampler and Hyperband pruning. The objectives are therefore to assess whether such a zonal MLP, fed only with intra-zonal and adjacent-zone counts plus simple marginal summaries, can recover both the spatial structure and global magnitude of OD demand in a highly underdetermined setting, to evaluate the effectiveness of the composite loss design under an optimized training configuration, and to study robustness by randomly removing a growing share of link flows before zonal aggregation and retraining for each data-availability scenario.

The remainder of the paper is organized as follows. Section 2 reviews classical and neural OD estimation methods together with relevant hyperparameter optimization literature and positions the present work within this context. Section 3 describes the simulation setup, zonal aggregation, input–output representation, neural architecture, hyperparameter optimization procedure, and loss functions. Section 4 presents the final model results and the sensitivity analysis under different levels of link-data removal. Section 5 discusses the implications and limitations of the findings, and Section 6 concludes with the main contributions and directions for future research.

2 Literature Review

Estimating the origin–destination (OD) matrix from traffic counts is a classic and challenging problem because many different OD patterns can reproduce the same set of observed link volumes. Early work by Robillard Robillard (1975) cast OD estimation as a linear regression problem under proportional assignment, often complemented by gravity-type models to distribute trips once costs are known. Van Zuylen and Willumsen Van Zuylen and Willumsen (1980) introduced information-minimization and entropy-maximization formulations that seek the most likely OD matrix consistent with observed flows, with or without a prior. Bell Bell (1983) refined these entropy-based models by iteratively adjusting a prior matrix to better fit link counts.

Probabilistic and statistical approaches subsequently added a more explicit treatment of uncertainty. Maher Maher (1983) formulated OD estimation in a Bayesian framework, treating both prior OD values and link counts as random variables and updating the prior via posterior inference. Spiess Spiess (1987) proposed maximum-likelihood estimators in which OD samples and, in an extension, link volumes are modeled as Poisson variables. Cascetta Cascetta (1984) introduced a Generalized Least Squares (GLS) method that combines survey-based start matrices with traffic counts while weighting discrepancies by their reliability; Bell Bell (1991) extended this with constrained GLS and variance expressions, improving applicability to larger networks.

In parallel, researchers explored formulations that more tightly couple OD estimation with traffic assignment and sensor placement. Yang Yang et al. (1994) showed that equilibrium-based bilevel OD-estimation models can often be transformed into single convex programs, clarifying their relationship to underspecified linear systems. Yang and Zhou Yang and Zhou (1998) analyzed how the number and location of counts affect identifiability and proposed optimization-based rules for sensor siting. Dynamic formulations emerged to capture temporal variation: Cascetta et al. Cascetta et al. (1993) developed dynamic simultaneous and sequential GLS-type estimators, while Ashok and Ben-Akiva Ashok and Ben-Akiva (2000) introduced a state-space and Kalman-filter framework for real-time estimation and short-term prediction. Zhou and Mahmassani Zhou and Mahmassani (2006) further integrated AVI data, link counts, and historical demand in a nonlinear bilevel formulation for online applications. Hazelton Hazelton (2000) proposed a Poisson-based likelihood method for uncongested networks, deriving feasible estimators via multivariate normal approximations. Castillo et al. Castillo et al. (2008) combined a Bayesian-network OD estimator with a Wardrop–minimum-variance assignment model and proposed correlation-based rules for counter placement.

More recently, advances in computational intelligence have motivated neural-network approaches that directly approximate the nonlinear mapping from traffic counts to OD flows. Kim and Chang Kim and Chang (2008) used a multilayer feed-forward network trained by backpropagation and reported improved accuracy over assignment-based methods such as MPME, since

the network bypasses repeated traffic loading. Mussone and Matteucci Mussone and Matteucci (2013) introduced a PCA-based preprocessing step to reduce dimensionality and stabilize training. Pamula and Zochowska Pamuła and Żochowska (2023) applied deep neural architectures to OD estimation and prediction in uncongested urban networks, showing that deep models can handle noisy or incomplete data. Min et al. Min et al. (2025) extended this idea to a digital-twin setting, using a deep network to exploit spatial and temporal correlations for continuous OD updates from real-time link counts; however, their model estimates link flows from known OD matrices, whereas the present work addresses the inverse problem of recovering OD demand from aggregated flow observations.

Although neural-network models provide a flexible and data-driven alternative to traditional OD estimation methods, their performance depends strongly on the selection of hyperparameters that control architecture, regularization, and training dynamics. These include quantities such as learning rate, batch size, hidden-layer width, dropout rate, and weight decay, which are not learned directly during optimization but must be specified before training Bergstra and Bengio (2012); Snoek et al. (2012). Early approaches often relied on manual trial-and-error or grid search, but such strategies become computationally inefficient in high-dimensional search spaces, especially when only a subset of hyperparameters has a strong influence on final model performance Bergstra and Bengio (2012); Probst et al. (2019).

To address this limitation, sequential model-based optimization methods have been developed to guide the search process more efficiently. Among these, the Tree-structured Parzen Estimator (TPE) is particularly suitable for neural-network pipelines with mixed and conditional hyperparameter spaces, since it can model promising and non-promising configurations separately and use this information to suggest improved candidate settings Bergstra et al. (2011). In addition, because neural-network training is computationally expensive, multi-fidelity approaches such as Hyperband allocate limited resources to many candidate configurations and prune weak trials early, thereby reducing overall search cost Li et al. (2018a). Asynchronous variants such as ASHA further extend this idea to parallel settings by allowing configurations to be promoted or stopped without synchronization barriers, improving scalability in practical tuning workflows Li et al. (2018b). Hybrid methods such as BOHB combine model-based search and resource-aware pruning to improve both efficiency and robustness in deep-learning optimization tasks Falkner et al. (2018). Population-Based Training (PBT) extends this perspective by allowing hyperparameters to evolve during training, making it possible to optimize schedules rather than only fixed values Jaderberg et al. (2017).

Recent studies also emphasize that neural-network performance depends not only on the search algorithm itself, but also on the choice of defaults, the structure of the search space, and the trade-offs embedded in the objective function. Holzmüller et al. Holzmüller et al. (2024) showed that carefully pre-tuned MLP configurations can substantially improve performance in tabular regression settings, highlighting the importance of strong defaults for structured prediction tasks. Probst et al. Probst et al. (2019) further showed that the gains from tuning are not distributed uniformly across hyperparameters, which supports prioritizing the most influential variables under limited computational budgets. In settings with composite objectives, hyperparameter selection can also be viewed as a trade-off problem among multiple criteria. Schmucker et al. Schmucker et al. (2020) demonstrated that multi-objective and multi-fidelity hyperparameter optimization provides a principled way to balance competing goals while still controlling computational cost.

These ideas are especially relevant in the present study, where the model includes not only

architectural and optimization hyperparameters, but also several interacting loss components used to preserve OD structure, marginal consistency, diagonal flows, and total-demand behaviour. In such settings, model performance depends not only on the neural architecture itself but also on how the hyperparameter configuration balances predictive accuracy, regularization, and training stability. For this reason, hyperparameter optimization is treated here as an integral part of the learning framework rather than as a separate post hoc step.

Over the past five decades, OD estimation has thus evolved from deterministic proportional methods to statistically grounded formulations and, more recently, to data-driven deep-learning models. Classical approaches emphasize theoretical consistency and explicit modeling of behavior, but can be demanding to calibrate and difficult to deploy in real time. Neural methods relax strong modeling assumptions and can capture complex nonlinear dependencies, yet often operate directly on high-dimensional link-flow vectors, pay limited attention to compact zonal representations, and rarely integrate systematic hyperparameter optimization into the overall estimation framework. The present study contributes to this emerging line of work by exploring a zonal, conservation-aware neural framework with automated hyperparameter optimization and by systematically assessing its robustness to information loss in link-level data.

3 Methodology

3.1 Input–Output Representation and Data Preparation

Each training sample corresponds to a short time window of simulated traffic in the Manhattan network, aggregated into K traffic analysis zones using K-means clustering. For every time window, a pair (X, Y) is constructed, where X represents observed zonal traffic counts and Y is the corresponding origin–destination (OD) demand matrix.

The input X does not contain flows between all zone pairs. Instead, it captures zonal traffic-count information derived from the underlying network and organized at the zone-to-zone level. In practice, the zonal count matrix is constructed from edge-based traffic counts and then used to represent the observed traffic pattern for each time window. This matrix is flattened and augmented with three summary features: origin totals (row sums), destination totals (column sums), and intra-zonal flows (diagonal terms). Depending on the selected configuration, the model can use either the full zonal count matrix together with these summary features or the summary features alone. The final input vector is:

$$\mathbf{x} = \big[\mathrm{vec}(X),\ \mathbf{o},\ \mathbf{d},\ \mathrm{diag}(X)\big] \in \mathbb{R}^{K^2+3K}.$$

The output $Y \in \mathbb{R}^{K\times K}$ is the true OD demand matrix obtained from the simulator. It is flattened into a vector $\mathbf{y} = \mathrm{vec}(Y)$ and transformed using $\log(1+Y)$ before being used as the learning target.

3.2 Train–Validation–Test Splitting and Temporal Chunking

To ensure fair temporal and level-wise comparison, a fixed splitting strategy is applied. For each congestion level, only the first 12 time slots (each 5 minutes) are retained and deterministically divided into:

$$8\ \text{(Train)},\quad 2\ \text{(Validation)},\quad 2\ \text{(Test)}.$$

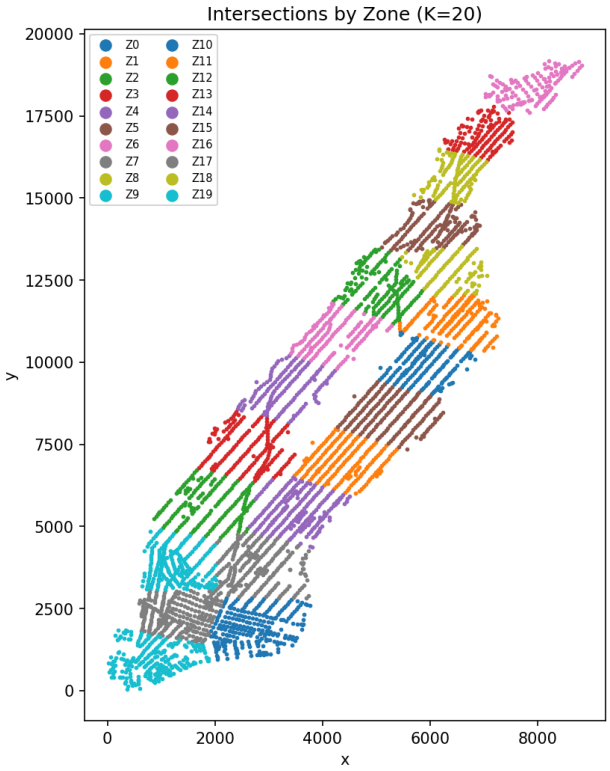


Figure 1: Intersections by zone for the Manhattan network ($K = 20$ K-means zones).

This guarantees that all congestion levels contribute equally to each subset and prevents temporal leakage.

To reduce noise and enable analysis at coarser time scales, consecutive time slots may be grouped into chunks of size c. For each chunk, both the input and OD matrix are aggregated by summation:

$$X^{(\text{chunk})} = \sum_t X^{(t)}, \quad Y^{(\text{chunk})} = \sum_t Y^{(t)}.$$

This fixed-slot design ensures that the train, validation, and test subsets remain comparable in duration and scale across congestion levels.

3.3 Feature Transformations

To improve numerical stability, both inputs and outputs are transformed. Depending on the selected configuration, the input features may first be transformed as:

$$\mathbf{x}' = \log(1 + \mathbf{x}),$$

and subsequently standardized using a scaler fitted only on the training set in order to avoid information leakage.

The target OD matrix is flattened and transformed similarly:

$$\mathbf{y} = \log(1 + Y),$$

and inverted at inference time using $Y = \exp(\mathbf{y}) - 1$.

3.4 Multi-Layer Perceptron Architecture

The OD estimation problem is formulated as a supervised regression task using a feed-forward Multi-Layer Perceptron (MLP) that takes as input the feature vector $\mathbf{x}$ defined in Section 3.1. Rather than fixing the architecture entirely *a priori*, the model is selected from a family of candidate MLP configurations through the hyperparameter optimization procedure described later in this section. In all cases, the network is composed of fully connected hidden layers whose widths are treated as tunable hyperparameters.

Each hidden layer is followed by batch normalization, ReLU activation, and dropout. In addition, a residual projection from the input layer to the final hidden representation is used to improve gradient flow and training stability. Depending on the selected configuration, the model may also include a separate output head for total-flow prediction.

The output layer predicts the flattened OD matrix in log-space to handle the large dynamic range of demand values.

3.5 Training Objective and Loss Function

The model is trained using a composite loss that separately constrains the *spatial structure* and the *global magnitude* of the OD matrix. The task is decomposed into learning: (i) the normalized spatial shape of demand and (ii) the total traffic volume.

The OD shape is defined as:

$$S = \frac{Y}{\sum_{i,j} Y_{ij}},$$

and the network predicts $\hat{S}$. The main training term is a hybrid shape loss:

$$\mathcal{L}_{\text{shape}} = 0.7\,\mathcal{L}_{L1}(S, \hat{S}) + 0.3\,\mathcal{L}_{Huber}(S, \hat{S}),$$

with the use of Huber loss treated as a configurable component within the final training framework. When enabled, the loss is computed with magnitude-based weighting to emphasize high-demand OD pairs.

Several auxiliary losses are introduced for spatial coherence. A **marginal loss** enforces consistency between predicted and true origin/destination totals. A **diagonal loss** improves estimation of intra-zonal flows, while a **percentage loss**, based on a symmetric MAPE formulation, controls relative errors across OD pairs with different magnitudes. These terms act as spatial regularizers.

When enabled within the selected configuration, the total traffic volume is predicted by a dedicated output head and constrained through the log-scale loss:

$$\mathcal{L}_{\text{total}} = \left(\log(T+1) - \log(\hat{T}+1)\right)^2, \quad T = \sum_{i,j} Y_{ij}.$$

A log-space reconstruction loss is further used to stabilize optimization across heterogeneous flow magnitudes. The full objective is:

$$\mathcal{L} = \mathcal{L}_{\text{shape}} + \lambda_{\text{marg}}\mathcal{L}_{\text{marg}} + \lambda_{\text{total}}\mathcal{L}_{\text{total}} + \lambda_{\text{diag}}\mathcal{L}_{\text{diag}} + \lambda_{\text{pct}}\mathcal{L}_{\text{pct}} + \lambda_{\text{log}}\mathcal{L}_{\text{log}}.$$

The coefficients $\lambda_{\text{marg}}, \lambda_{\text{total}}, \lambda_{\text{diag}}, \lambda_{\text{pct}}$, and λ_{log} are treated as tunable hyperparameters within the optimization procedure.

Table 1: Loss terms and their interpretation

Loss Term	Formula	Interpretation
$\mathcal{L}_{shape}$	$0.7\|S-\hat{S}\|_1 + 0.3\,\mathcal{L}_{Huber}(S,\hat{S})$	Main spatial pattern loss
$\mathcal{L}_{marg}$	$\|\hat{S}\mathbf{1}-S\mathbf{1}\|^2 + \|\mathbf{1}^T\hat{S}-\mathbf{1}^TS\|^2$	Origin and destination marginal consistency
$\mathcal{L}_{diag}$	$\|\mathrm{diag}(\hat{S})-\mathrm{diag}(S)\|_2^2$	Penalizes errors on internal (diagonal) flows
$\mathcal{L}_{pct}$	$\frac{\|\hat{S}-S\|}{\|\hat{S}\|+\|S\|+\varepsilon}$	Relative (percentage) OD error control
$\mathcal{L}_{total}$	$(\log(T+1)-\log(\hat{T}+1))^2$	Total demand consistency
$\mathcal{L}_{log}$	$\|\log(1+\hat{Y})-\log(1+Y)\|_2^2$	Log-space reconstruction stability

3.6 Hyperparameter Optimization

Because the performance of neural networks depends strongly on architectural, optimization, regularization, and loss-related hyperparameters, model selection was performed through a systematic hyperparameter optimization procedure rather than manual trial-and-error. The search was implemented in Optuna using a Tree-structured Parzen Estimator (TPE) sampler together with Hyperband pruning in order to efficiently explore a high-dimensional search space while terminating weak trials early.

For each trial, a candidate hyperparameter configuration was sampled and passed to the main model-training script. The objective value was defined as the best validation sMAPE achieved during training, and the optimization problem was formulated as a minimization task. Intermediate validation results were reported periodically during training so that unpromising trials could be pruned before consuming the full computational budget.

The search space included architectural hyperparameters such as hidden-layer configuration and dropout rate; optimization and training hyperparameters such as learning rate, weight decay, batch size, maximum number of epochs, and early-stopping patience; loss-related hyperparameters such as the weights associated with marginal, diagonal, percentage, log-space, and total-flow terms; and optional stabilization or post-processing components including Huber loss settings, magnitude weighting, total-flow prediction, iterative proportional fitting (IPF), global rescaling, and related control parameters. After completion of the search, the configuration yielding the lowest validation sMAPE was selected as the final model setting used for all reported results.

Formally, the hyperparameter search can be expressed as:

$$\theta^* = \arg\min_{\theta\in\Theta} \ \mathrm{sMAPE}_{\mathrm{val}}(\theta),$$

where Θ denotes the hyperparameter search space and $\mathrm{sMAPE}_{\mathrm{val}}(\theta)$ is the best validation sMAPE obtained by the model under configuration θ.

3.7 Optimization, Early Stopping, and Post-processing

Model parameters are optimized using the AdamW optimizer with a validation-based learning rate scheduler. The learning rate is reduced automatically when validation improvement plateaus.

Early stopping is applied based on validation sMAPE. Training is terminated if no improvement is observed within a fixed patience window, and the model with the best validation performance is retained. During hyperparameter optimization, intermediate validation results are also reported periodically so that unpromising trials can be pruned before consuming the full computational budget.

Several optional stabilization and post-processing components are considered within the final framework. These include a separate total-flow prediction head, global rescaling, gamma-based sharpening, and iterative proportional fitting (IPF)-based correction steps intended to improve structural consistency in the reconstructed OD matrix. Their inclusion depends on the selected hyperparameter configuration.

4 Results

4.1 Model Performance Before and After Hyperparameter Optimization

The proposed MLP-based model was first evaluated using the original baseline configuration and was then re-evaluated after incorporating hyperparameter optimization into the training pipeline. The purpose of this comparison is to highlight the effect of tuning on the predictive performance of the same overall OD-estimation framework, while keeping the remainder of the results presentation unchanged.

Before hyperparameter optimization, the model achieved sMAPE values of 13.16%, 22.94%, and 21.56% on the training, validation, and test sets, respectively. After tuning, these values improved to 10.82%, 16.73%, and 16.07%, indicating a clear reduction in reconstruction error across all three splits. The largest gains were observed on the validation and test sets, suggesting that the tuned configuration improves generalization in addition to fitting the training data more effectively.

In terms of total demand reconstruction, the relative error in predicted total traffic volume also decreased after tuning. The total-flow error changed from 4.79%, 4.80%, and 3.95% to 3.33%, 4.69%, and 1.70% across the training, validation, and test sets, respectively. Although marginal MAE increased after tuning, the overall improvement in sMAPE and total-flow consistency suggests that hyperparameter optimization strengthened the model's ability to recover the global OD structure while maintaining strong predictive performance on unseen data.

Table 2: Performance metrics before and after hyperparameter optimization.

Metric	Model	Train	Validation	Test
sMAPE (%)	Before tuning	13.16	22.94	21.56
	After tuning	10.82	16.73	16.07
MAE	Before tuning	3.55	3.07	2.89
	After tuning	3.35	3.15	3.04
RMSE	Before tuning	4.52	3.92	3.76
	After tuning	4.45	4.01	3.94
Marginal MAE	Before tuning	27.98	15.11	14.46
	After tuning	32.01	17.88	16.63
Total Flow Error (%)	Before tuning	4.79	4.80	3.95
	After tuning	3.33	4.69	1.70

Table 3: Training OD matrix for demand level 11,756 (20 zones) – Training set

Zone	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	45	31	30	7	22	26	24	30	26	40	28	22	25	31	35	29	35	40	18	38
2	19	15	20	11	9	14	18	20	11	25	15	24	17	9	29	26	9	28	7	23
3	18	10	16	15	15	18	14	23	7	31	12	23	9	26	22	25	18	15	10	16
4	16	17	19	11	11	15	8	17	8	12	13	16	13	8	22	7	15	17	5	14
5	17	19	16	9	17	9	17	14	6	17	18	8	16	14	22	7	5	32	3	10
6	32	18	23	7	13	25	18	15	19	23	14	17	13	14	17	16	14	23	9	14
7	15	15	14	8	10	18	13	13	9	23	8	12	20	22	13	10	17	20	15	7
8	27	14	22	22	15	22	20	37	14	25	28	20	23	25	26	16	21	26	13	46
9	15	12	11	5	5	13	14	21	12	28	9	4	12	16	17	6	14	22	5	18
10	36	38	17	10	17	33	15	21	18	19	23	23	23	36	30	24	26	30	16	28
11	14	13	22	15	13	17	24	23	23	18	21	19	5	19	31	25	17	27	12	19
12	33	10	23	19	12	19	9	17	11	25	16	13	14	17	37	12	15	26	17	24
13	33	25	24	13	16	16	15	26	9	22	14	20	20	16	19	18	21	21	3	26
14	23	22	12	15	16	21	10	23	13	30	18	17	13	18	28	18	18	24	11	19
15	40	22	15	11	25	29	22	18	12	30	16	28	24	15	40	32	21	41	11	29
16	24	18	20	11	13	27	21	29	14	21	23	23	10	27	28	21	8	23	19	39
17	13	17	13	13	6	16	14	11	10	22	13	15	13	16	24	16	15	21	11	26
18	44	28	28	28	17	28	24	24	19	35	24	30	20	33	24	28	28	20	17	30
19	17	10	9	7	11	7	5	10	13	20	13	14	17	9	11	17	8	18	3	14
20	39	19	23	21	15	22	19	29	21	24	22	17	23	19	31	24	7	30	16	28

Table 4: Predicted OD matrix for demand level 11,756 (20 zones) – Training set

Zone	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	43	23	33	10	21	29	20	24	25	24	27	22	29	28	42	27	27	35	10	35
2	26	21	18	14	14	15	20	19	11	20	16	24	15	19	22	24	13	22	3	19
3	25	13	14	17	11	22	16	25	9	24	13	20	5	18	21	22	12	17	11	22
4	19	18	18	11	9	10	9	15	11	14	10	16	9	12	20	9	18	21	8	13
5	18	17	13	11	15	11	15	11	4	18	7	11	13	15	25	10	6	25	6	14
6	28	19	15	13	16	15	14	20	11	14	15	16	15	16	20	25	14	25	13	18
7	18	16	13	11	10	12	16	13	10	16	11	13	19	16	5	15	15	18	16	6
8	25	17	22	15	14	24	20	32	16	19	23	20	25	20	24	16	17	31	16	42
9	16	12	10	10	6	9	13	16	9	25	10	10	10	18	21	6	13	23	9	22
10	33	28	23	14	18	31	15	28	19	21	17	21	27	39	27	20	19	29	13	23
11	15	16	18	17	13	18	17	20	17	20	16	18	17	22	18	24	18	30	11	18
12	30	15	18	19	13	15	9	14	13	33	10	12	18	19	29	12	15	25	13	24
13	30	24	24	14	16	14	11	24	9	24	15	12	18	15	17	17	25	18	9	26
14	23	16	12	10	15	16	14	27	15	25	17	20	9	19	20	16	17	26	5	23
15	28	24	22	11	22	25	19	22	10	33	19	25	22	22	34	26	16	37	16	28
16	27	20	14	5	16	23	16	25	14	22	18	19	15	23	35	20	10	27	15	34
17	15	15	11	8	10	13	16	12	14	25	15	13	13	16	21	15	15	24	9	21
18	41	31	27	25	16	31	23	25	18	40	24	30	22	0	23	26	24	20	18	32
19	5	10	11	9	7	5	7	12	8	14	11	12	20	13	16	10	11	16	4	14
20	31	25	24	18	16	18	22	10	20	32	23	23	25	18	20	25	14	28	15	24

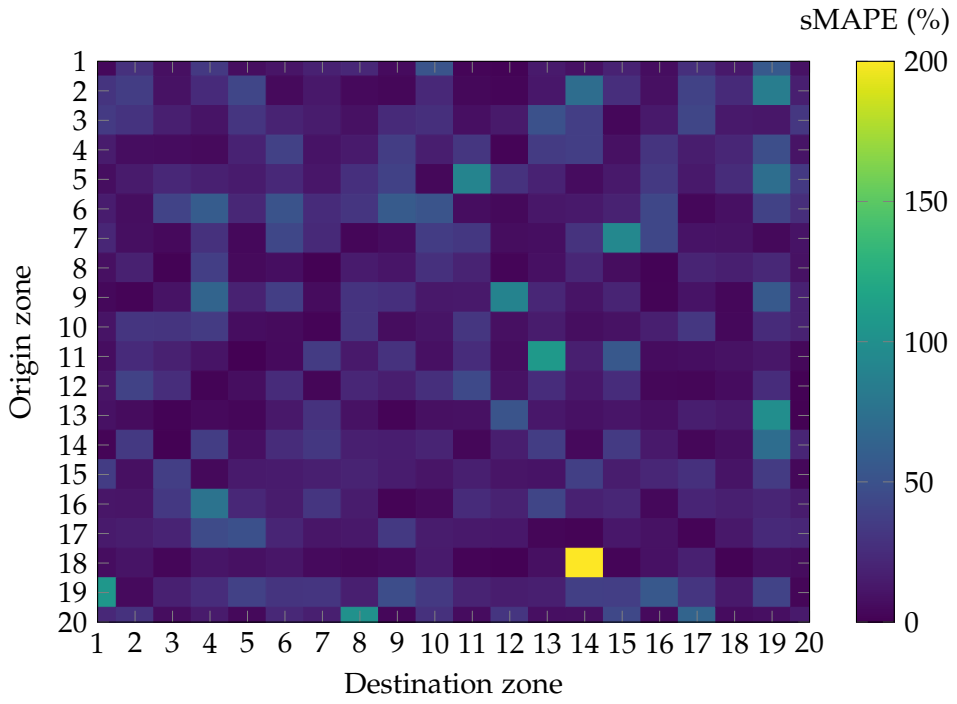


Figure 2: Cell-wise sMAPE (%) between true and predicted OD flows – level 11,756 (20 zones), training set

4.2 Sensitivity to Reduced Input Information

A sensitivity analysis was conducted to evaluate the impact of reduced input information on model performance. For each scenario, a fixed proportion p (from 0% to 99.5%) of individual link flows was randomly selected and removed from the underlying network data. The zonal matrix X, containing intra- and inter-zonal traffic counts, was then recomputed from the remaining link flows by aggregating internal links (within each zone) and boundary links (between adjacent zones). The resulting degraded matrices X were used as inputs, and the model was retrained under identical settings for each value of p. Performance was assessed on the corresponding test sets.

Three indicators are reported: sMAPE to quantify cell-level OD accuracy, total flow error to measure global traffic consistency, and marginal MAE to capture errors in origin–destination totals. Under moderate information loss (up to 60% masking), the test sMAPE remains relatively stable, staying close to the baseline value of 24.23%. At the same time, the total flow error increases only slightly, remaining below 10% for most cases, indicating that the model preserves a reasonable estimation of global demand even when a significant portion of link-level information is missing.

However, when more than 80% of the input information is removed, performance degrades more clearly. The sMAPE increases beyond 25%, reaching 30.04% at 99.5% masking. In parallel, the total flow error rises sharply and exceeds 18% at extreme masking levels, reflecting the model's reduced ability to maintain global demand consistency. Moreover, marginal MAE also increases substantially, indicating growing inconsistencies in predicted origin and destination totals. These results confirm that the model remains robust under partial data loss but becomes

less reliable when spatial traffic information is severely limited.

Table 5: Model performance under different levels of input data removal

Available Data (%)	sMAPE (%)	Marginal MAE	Total Flow Error (%)
100	24.23	15.72	2.85
80	23.84	17.04	4.01
60	23.63	22.20	9.22
40	23.39	18.28	6.82
20	24.25	18.53	7.95
10	25.04	25.19	13.42
5	25.15	22.25	10.93
3	26.78	24.27	10.61
2	28.37	37.94	16.72
1	28.39	36.00	20.90
0.5	30.04	50.45	29.53

5 Discussion

The results show that an MLP fed only with intra-zonal and adjacent-zone counts, plus simple marginal summaries, can reconstruct full OD matrices with moderate cell-wise errors and small total-flow discrepancies. This indicates that the dominant spatial structure of demand, together with its global magnitude, can be inferred from a compact zonal representation constructed from link data.

The comparison between the original baseline configuration and the hyperparameter-optimized version further shows that tuning improves the predictive performance of the same overall framework. In particular, the tuned model reduced sMAPE across the training, validation, and test sets, with the most visible gains observed on validation and test data. This suggests that the hyperparameter optimization procedure improved not only fit to the training set, but also the generalization ability of the model. The reduction in total-flow error after tuning, especially on the test set, also indicates that the optimized configuration better preserves the global magnitude of OD demand.

The cell-wise error analysis highlights that discrepancies are concentrated in low-flow and highly variable OD pairs, while most medium- and high-flow relations are reconstructed with moderate error levels. This is consistent with the behaviour of sMAPE, which penalizes relative errors more strongly for small values. From an operational perspective, the fact that major OD corridors are better reconstructed is encouraging, as these flows typically drive network performance and congestion dynamics.

The sensitivity analysis with respect to reduced input information provides further insight into robustness. When up to about 60% of link flows are removed prior to zonal aggregation, performance degradation remains limited, suggesting that the model exploits redundant spatial information and learned demand regularities rather than depending on individual sensors. For very high levels of data loss (above 80%), however, both sMAPE and total-flow errors increase substantially, indicating the existence of a minimum information threshold below which the OD reconstruction problem becomes too ill-posed even for a data-driven model.

Several limitations of the present study must be acknowledged. The model was trained and evaluated exclusively on synthetic data generated for only four traffic demand levels over a total duration of four hours. As a result, the variability of traffic conditions, demand patterns, and network states is limited compared to real-world scenarios, and the reported performance should be interpreted as an upper-bound estimate under controlled conditions. In addition, the zoning and network configuration remain fixed throughout the analysis, and the generalization of the approach to other network structures or different spatial aggregations has not yet been tested. Despite these limitations, the results demonstrate the potential of neural OD estimation methods as efficient surrogates for traditional optimization-based approaches, and also show that systematic hyperparameter optimization can further strengthen their predictive performance.

6 Conclusion

This paper proposed a supervised deep-learning framework that reconstructs origin–destination (OD) matrices from sparse zonal traffic counts in a 20-zone Manhattan network. The multilayer perceptron (MLP) takes as input intra-zonal flows and flows between adjacent zones, together with simple marginal summaries derived from link counts, and recovers full OD matrices on synthetic data with moderate cell-wise errors and small total-flow discrepancies.

Relative to existing neural approaches Kim and Chang (2008); Mussone and Matteucci (2013); Pamuła and Żochowska (2023); Min et al. (2025), the contribution is fourfold. First, the model operates on a compact zonal representation constructed from link flows, aggregating internal and boundary links into intra-zonal and adjacent-zone traffic counts, which reduces input dimensionality while preserving physical interpretability. Second, the learning objective explicitly distinguishes OD shape from scale and augments a standard regression loss with marginal, diagonal, and total-flow consistency terms, providing a principled way to encode basic conservation-of-flow properties during training. Third, a systematic sensitivity analysis quantifies how performance degrades as link-based information is progressively removed, highlighting that the model remains reasonably robust under moderate data loss but deteriorates when spatial coverage becomes extremely sparse. Fourth, the incorporation of hyperparameter optimization further improves the performance of the baseline framework, particularly in terms of sMAPE and total-flow reconstruction, demonstrating that tuning plays an important role in extracting stronger performance from the same core neural architecture.

The study is subject to several limitations, notably the reliance on synthetic data over a limited time horizon and a single zoning scheme. Future work will focus on validation with real detector data, longer and more heterogeneous periods, alternative zoning and network structures, and the integration of graph-based neural architectures and online updating within a digital-twin framework. Future work could also explore robustness-oriented training strategies so that performance remains stronger under severe information loss. Overall, the results indicate that carefully designed deep-learning models can serve as practical surrogates for OD estimation in settings with sparse and noisy traffic measurements, and that systematic hyperparameter optimization provides an effective way to improve their predictive accuracy.

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